

How To Employ ORGANIC PHOTOVOLTAIC SOLAR CELLS For Energy Conversion

In the last few years, performance of organic solar cells has significantly improved. Further improvements are needed for the further development of organic photovoltaic cells. This article reviews the basic working principle, commercialisation and the fate of organic solar cells. Importance of high power conversion efficiency models for organic solar cells and the future development in the field of nanotechnology are also discussed



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Today, we are all aware of the fact that fossil fuels such as natural gas, coal and oil are finite resources available for a finite time. Excessive worldwide usage of these resources has introduced conflicts and such modern concepts as global warming with the consequence of the depleting ozone layer. And total demand for energy is increasing day by day.

An organic solar cell is a type of photovoltaic that uses organic electronics—a branch that deals with conductive organic polymers or small organic molecules for light absorption and charge transport to produce electricity from sunlight using photovoltaic effect. Organic solar cells are cheap, resulting in low production costs to form large volumes. Due to the high optical absorption coefficient (α) of organic molecules, large amounts of sunlight can be absorbed easily and efficiently.

Compared to conventional silicon solar cells, organic solar cells are lighter in weight, easily disposable, convenient to fabricate, potentially transparent, flexible and come with a lower manufacturing cost.

After considering the above merits, organic solar

cells find wider applications than conventional ones. Organic photovoltaics can be installed on windows or rooftops, as shown in Fig. 3.

Fig. 4 shows transparent solar organic solar cells designed and manufactured by a German company called BASF. The firm produces creative organic solar organic photovoltaics and has published several researches in the field.

BASF has successfully manufactured transparent organic bulk hetero junction solar cells, and is now trying to expand this activity.

The physics of organic photovoltaics

Actual structure of an organic photovoltaic is made of two materials: fullerene

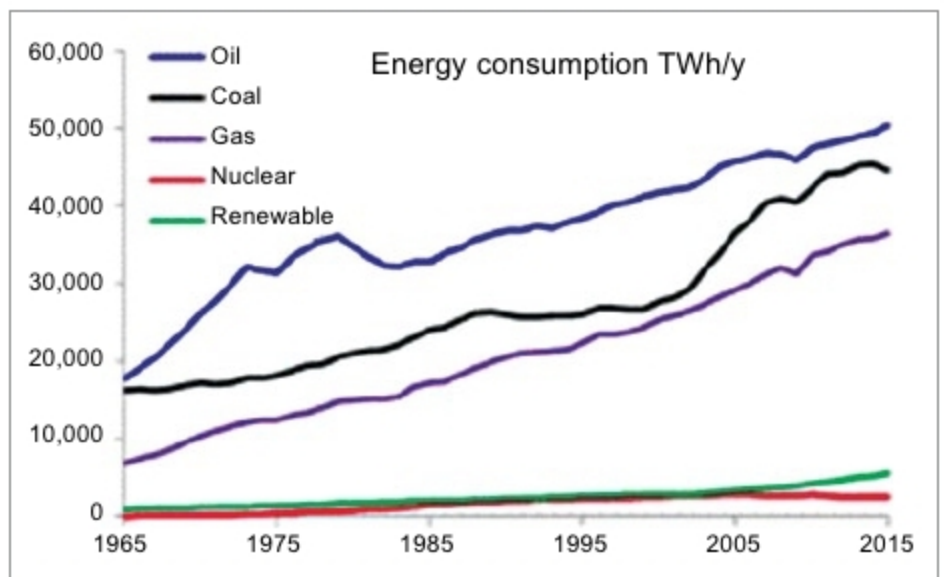


Fig. 1: Global statistical utilisation scenario of energy sources